

1 Homework

1. Suppose $(X, \mathcal{T})$ is a topological space. Let $K \subseteq X$ be compact, and let $f : K \rightarrow \mathbb{R}$ be continuous. Show that there exist $x_1, x_2 \in K$ such that for every $x \in K$,

$$f(x_1) \leq f(x) \leq f(x_2).$$

2. Show that if a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x) = 0$ for all $x \in \mathbb{Q}$, then $f(x) = 0$ for all $x \in \mathbb{R}$.
3. Give an example of a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ and an open set U such that $f(U)$ is not open.
4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Suppose that a sequence of functions $f_n(x)$ on $\mathbb{R}$ satisfy:
- (a) $0 \leq f_n(x) \leq 1$ for all $x \in \mathbb{R}$ and $n \geq 1$.
 - (b) $f_n(x)$ is increasing in x for every $n \geq 1$.
 - (c)

$$\lim_{n \rightarrow \infty} f_n(x) = f(x)$$

for each $x \in \mathbb{R}$.

(d)

$$\lim_{x \rightarrow -\infty} f(x) = 0, \quad \lim_{x \rightarrow \infty} f(x) = 1.$$

Show that $f_n(x) \rightarrow f(x)$ uniformly on $\mathbb{R}$.

5. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$|f(x) - f(y)| \leq M|x - y|$$

for all $x, y \in \mathbb{R}$. Such a function is called Lipschitz. Suppose that f is differentiable.

- (a) Show that f is continuous.
- (b) Show that

$$|f'(x)| \leq M$$

for every $x \in \mathbb{R}$.

- (c) Conversely, suppose that

$$|f'(x)| \leq M$$

for all $x \in \mathbb{R}$. Show that

$$|f(x) - f(y)| \leq M|x - y|$$

for all $x, y \in \mathbb{R}$.

6. Prove that

$$|\sin x - \sin y| \leq |x - y|.$$

7. Suppose f is k -times differentiable on an open interval I and

$$f^{(k)}(x) = 0$$

for all $x \in I$. What form does f have? Prove your claim.

8. Let

$$f(x) = \begin{cases} x, & \text{if } x \text{ is irrational,} \\ 0, & \text{if } x \text{ is rational.} \end{cases}$$

- (a) Calculate the upper and lower Darboux integrals of f on the interval $[0, b]$.
- (b) Give another example of a bounded function that is not Riemann integrable.

9. Show that if f is integrable on $[a, b]$, then f is integrable on every interval $[c, d] \subseteq [a, b]$.

Discussion

Riemann Integration

1. Let f be a bounded function on $[a, b]$. Prove that f is integrable if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every partition P of $[a, b]$ with $\text{mesh}(P) \leq \delta$,

$$U(f, P) - L(f, P) < \varepsilon.$$

2. Prove that a bounded function f on $[a, b]$ is Darboux integrable if and only if it is Riemann integrable.

Cantor Set

1. Let

$$A_0 = [0, 1], \quad A_1 = [0, \tfrac{1}{3}] \cup [\tfrac{2}{3}, 1], \quad A_2 = [0, \tfrac{1}{9}] \cup [\tfrac{2}{9}, \tfrac{1}{3}] \cup [\tfrac{2}{3}, \tfrac{7}{9}] \cup [\tfrac{8}{9}, 1].$$

For all n , define

$$A_{n+1} = \frac{A_n}{3} \cup \left(\frac{2}{3} + \frac{A_n}{3} \right).$$

Set

$$C = \bigcap_{n=0}^{\infty} A_n.$$

We call C the Cantor set. Show:

- (a) C is compact.
- (b) C has infinitely many points. (Hint: Consider the endpoints of the intervals.)
- (c) $\text{int}(C) = \emptyset$.
- (d) C is totally disconnected; that is, given any $x, y \in C$, there is a disconnection U and V of C such that $x \in U$ and $y \in V$.
- (e) Every point of C is a cluster point of C (sets with this property are called perfect).